AXIS
PERFORMANCE SCIENCE

FREE GUIDE

Knee Pain Starter Guide

A two-week plan for pain at the front of the knee. Ten minutes a day, three strength sessions a week.

EXERCISE SCIENTIST

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BEFORE YOU START

Who this guide is for

Pain at the **front of the knee**, around or behind the kneecap. It is usually worse on stairs, squatting, running, or after sitting for a long time. This is common in active adults, and it is what the research below studied.

It is a starter, not a diagnosis. If your pain sits somewhere else, or any of the signs below apply, get the knee checked first.

See a doctor first if

- The knee gave way, locked, or you heard a pop when it started
- Swelling came on within hours, or the knee will not fully straighten
- The pain began after a direct blow, or a twist with the foot planted
- There is redness, heat or fever, or night pain that wakes you

This guide is general information for pain management and prevention. It does not replace a medical assessment.

WHAT THE RESEARCH SAYS

Train the hip as well as the knee

In people with front-of-knee pain, adding hip strengthening to knee strengthening worked better than knee strengthening alone.

NASCIMENTO ET AL., J ORTHOP SPORTS PHYS THER, 2018

14 trials, 673 people, pooled in a meta-analysis

Hip plus knee strengthening lowered pain by a further **1.5 points** than knee strengthening alone (95% CI 0.8 to 2.3). Activity also improved more, and the benefit lasted beyond the training period.

ZHANG ET AL., EUR J MED RES, 2025

12 studies, 1,066 people, pooled in a meta-analysis

Hip strengthening reduced pain more than knee strengthening alone, and both types helped.

What this does not tell you

- Many of the trials were small and of limited quality.
- Results are group averages. Your knee may respond differently.
- Pain improved even when measured strength barely changed, so the exact reason is still debated.

That is why this plan trains both: the hip above the knee and the muscles around it.

GROUND RULES

How much pain is okay?

Rehab does not have to be pain-free. It has to be **controlled**. Rate your pain from 0 to 10 during each set.

0–2	**Safe.** Train exactly as written.
3–5	**Acceptable.** Keep going if both rules below hold.
Above 5	**Stop.** Reduce range, then load, then switch to an easier version.

The two rules that matter more than the number

- Pain settles back to normal **by the next morning**.
- Your everyday pain is **trending down** across the two weeks.

Pain that lingers into the next day, or creeps up week after week, means the dose was too big, whatever the score was at the time. Do less next session.

How hard to push

Finish each set with about **3 to 4 reps left in the tank**. If the last rep looks the same as the first, the weight or the step height is right.

EVERY DAY · 10 MINUTES

Daily mobility

Do this every day. On strength days it doubles as your warm-up. Rest 30 seconds between moves. Tap any name for a video.

01 **Ankle dorsiflexion mobilization**

`2 × 8 per side`

Knee travels forward over the middle toes. Heel stays down.

02 **Deep squat hold**

`2 × 30 seconds`

Hold on to something for balance if you need to.

03 **Hip flexor stretch (lunge)**

`2 × 30 seconds per side`

Squeeze the glute of the back leg.

04 **Supine figure-4 stretch**

`2 × 30 seconds per side`

Never into sharp pain.

05 **Banded lateral walks**

`2 × 12 steps`

Feet stay wide, hips low, 2 seconds per step.

The non-negotiable: keep walking

20 to 30 minutes of easy walking on flat ground, most days. If walking hurts, shorten it and split it into two walks rather than stopping.

3 DAYS A WEEK · NOT BACK TO BACK

Strength sessions

Do the daily mobility first, then these six. About 25 minutes. A mat, a band, a towel and a low step are all you need.

01 **Glute bridge**

3 × 12–15 · rest 45 s

1 second up, 2 second squeeze, 2 seconds down. Drive through the heels.

02 **Clamshells**

3 × 15 per side · rest 45 s

Band above the knees. Pelvis stays stacked, no rolling back.

03 **Calf raise on a step**

3 × 12–15 · rest 60 s

Full range. 1 second up, 3 seconds down, pause at the bottom.

04 **Hamstring sliders**

3 × 8–10 · rest 60 s

Towel on a smooth floor. Hips stay up the whole set.

05 **Side plank**

3 × 20–30 seconds per side · rest 45 s

Body in one line. Hips do not sag.

06 **Forward step-downs**

2 × 8 per side · rest 60 s

Low 10–15 cm step. Tap the heel down slowly. Knee over the middle toes.

WEEK 2

How to progress

The second week is harder on purpose, so you are never guessing.

- Add **one set** to the glute bridge and the clamshells.
- Add **5 seconds** to every hold.
- Only move to a higher step or add load if the current version was clean at every set and the knee settled by the next morning.
- Not there yet? Repeat week 1. The calendar serves you, not the other way round.

Short on time?

Keep this priority order: daily mobility, glute bridge, step-downs, calf raises. Those build the capacity that protects the joint.

Adding load

Once the step-down is clean for two sessions in a row, add 2 to 2.5 kg in a backpack or a light dumbbell, or move to a slightly higher step.

CHECK YOUR PROGRESS

Two tests, five minutes

Knee-to-wall

Face a wall in a half-kneel, foot flat. Drive the knee forward to touch the wall without the heel lifting. Slide the foot back until the knee only just touches. Measure wall to big toe, each side. **Under about 10 cm** points to a stiff ankle, which loads the knee more.

Step-down, filmed from the front

10 slow reps each side from a 20–30 cm step. Does the knee stay over the middle toes, or fall inward? You cannot feel this, so film it.

Score sheet

Measure	Day 1	Day 14
Knee pain this week (0–10)		
Pain on stairs (0–10)		
Knee-to-wall, left (cm)		
Knee-to-wall, right (cm)		
Step-down: knee tracks or caves		

WHAT TO EXPECT

The first two weeks

Week 1–2

Stair pain and morning stiffness usually ease first. The knee-to-wall number starts to grow and the step-down looks steadier.

If nothing has moved by week 3

Do not push harder. Get a professional assessment from a physiotherapist or sports-medicine doctor.

Common mistakes

- **Chasing stretches and skipping load.** Stiffness feels like the problem, but weakness is often the cause.
- **Resting completely.** Rest settles the knee for a week, then the pain returns when you train again.
- **Going too hard on day one.** Next-day pain means the dose was too big.

Two free helpers

Sleep **7 to 9 hours** and manage stress. Both affect how sensitive pain feels.

WHAT'S NEXT

The full six-week program

This guide is the first two weeks of the AXIS Knee Pain Corrective Program. The full version goes further, in three phases:

01 **Control**
Weeks 1–2. Isolate and control. This guide.

02 **Build**
Weeks 3–4. Load the big patterns: leg press, step-downs, Romanian deadlift.

03 **Integrate**
Weeks 5–6. Single-leg strength, landing and change of direction.

Be first to know

AXIS programs are coming. Join the waitlist and you will hear first, with early access.

Link in bio at **@a_bouzubar**

SOURCES

References

Retrieved from PubMed.

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Zhang Z, Zhang Z, Zheng B, Yang Y, Sun Y. Effects of lower limb strengthening training on lower limb biomechanical characteristics and knee pain in patients with patellofemoral pain: a systematic review and meta-analysis. *Eur J Med Res.* 2025;30(1):90. doi:10.1186/s40001-025-02347-3

The pain guide, exercise selection and progression rules are AXIS applied practice built on this evidence, not a direct result from any single study.

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